

ASSIGNMENT #05

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2025
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Names : _____
Due Date: Thursday, October 30th my Office (06:00PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem: The following **virial** equation of state (originally developed by Beattie and Bridgeman, *Proc. Am. Acad. Arts Sci.*, **63**, 229-308 (1928)) was proposed by Deming and Shupe, *J. Am. Chem. Soc.*, **52**, 1382-1389 (1930), for the prediction of the thermal properties of Nitrogen in the temperature range $[-70^{\circ}\text{C}, 400^{\circ}\text{C}]$ and in the pressure range $[0\text{ atm}, 400\text{ atm}]$

$$PV^2 = RT \left[V + B_o \left(1 - \frac{b}{V} \right) \right] \left(1 - \frac{c}{VT^3} \right) - A_o \left(1 - \frac{a}{V} \right)$$

where $\{a, A_o, b, B_o, c\}$ are numerical constants that were determined from the experimental data.

The values reported by Deming and Shupe are the following when P the pressure, is in atm , V the specific volume, is in g/cm^3 and R the Universal Gas constant, in $\text{cm}^3 \cdot \text{atm} / (\text{g} \cdot \text{K})$

Table 1. Virial Equation Constants

$A_o = 1528.6$	$a = 0.2748$	$c = 2.2 \times 10^6$
$B_o = 1.643$	$b = -0.9235$	

Later in 1952, Lunbeck, Michels and Wolkers, *Appl. Sci. Res.*, A3: 197 (1952), reported Z the compressibility factor of Nitrogen at very high pressures. The data are shown in the table below:

P (atm)	Temperature (deg C)				
	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.999	1.000	1.001	1.001
10	0.996	0.998	1.000	1.001	1.002
30	0.959	0.996	1.001	1.005	1.007
50	0.958	0.996	1.004	1.009	1.014
100	0.984	1.004	1.018	1.027	1.033
200	1.036	1.057	1.072	1.082	1.089
300	1.134	1.146	1.154	1.160	1.163
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

The following modification of the Deming-Shupe **virial** equation is proposed to include the new high-pressure data:

$$PV^2 = RT \left[V + B_o \left(1 - \frac{b}{V} \right) \right] \left(1 - \frac{c}{VT^3} \right) - A_o \left(1 - \frac{a}{V} - \frac{d}{V^2} \right)$$

where d is a new numerical constant to be determined by the high-pressure data.

Problem #1

1. Develop a methodology to estimate the new constant d from the high-pressure experimental by minimizing the following objective function:

$$\Phi(d) = \min \sum_i (P_i^{\text{exp}} - P_i^{\text{model}})^2$$

Hint: Recast the virial equation of state in the form

$$Z = 1 + \beta\rho + \gamma\rho^2 + \delta\rho^3$$

where Z is the compressibility factor, ρ is the molar[or mass] density of nitrogen and $\{\beta, \gamma, \delta\}$ are coefficients that depend on temperature and on the constants $\{a, A_o, b, B_o, c, d\}$. For the estimation of the new constant d keep the values of the rest of the constants $\{a, A_o, b, B_o, c\}$ the same as given in the **Table 1** above. Prepare two **parity plots**, the first with the value of the constant d set equal to zero and the second with the optimal value calculated by the nonlinear regression.

In this problem we are given the virial equation $PV^2 = RT \left[V + B_o \left(1 - \frac{b}{V} \right) \right] \left(1 - \frac{c}{VT^3} \right) - A_o \left(1 - \frac{a}{V} - \frac{d}{V^2} \right)$

We need to simplify the equation into the form $Z = 1 + \beta\rho + \gamma\rho^2 + \delta\rho^3$, and then once we have the virial equations in forms of Z we can create a vba function based off our density, beta, gamma, and delta values that will calculate Z values for us. From our textbook we know that we can make Z a function of density (ρ) and Temperature (T) through the relationship $Z = \frac{PV}{RT}$, which means that $V = \frac{ZRT}{P}$ and because Volume and density are inversely proportional $\rho = \frac{P}{ZRT}$.

Derivation of β, γ, δ :

We started by looking at the left side of the equation by looking at one term of the virial equation first and that term is $V + B_o \left(1 - \frac{b}{V} \right)$. We expanded the expression and multiplied B_o by our terms in the parentheses and turned the equation into the form $V + B_o \left(1 - \frac{b}{V} \right)$.

The next step is to continue working through the equation and multiply by the following term $1 - \frac{c}{VT^3}$

$$\left(1 - \frac{c}{VT^3} \right) \left(V + B_o \left(1 - \left(\frac{b}{V} \right) \right) \right) = V + B_o - \frac{B_o b}{V} - \frac{c}{T^3} - \frac{B_o c}{VT^3} + \frac{B_o b c}{V^2 T^3}$$

The next step in the virial equation is multiplying our expanded term by RT which gives us

$$RT \left(V + B_o - \frac{B_o b}{V} - \frac{c}{T^3} - \frac{B_o c}{VT^3} + \frac{B_o bc}{V^2 T^3} \right) = VRT + RTB_o - \frac{RTB_o b}{V} - \frac{RC}{T^2} - \frac{RB_o C}{VT^2} + \frac{RB_o bc}{V^2 T^2}$$

Then we looked at the term $-A_o \left(1 - \frac{a}{Z} - \frac{d}{V^2} \right)$ and expanded it into $-A_o + \frac{A_o a}{Z} + \frac{A_o d}{V^2}$. Now we can combine all the terms of the left side of the equation

$$PV^2 = RTV + RTB_o - \frac{RTB_o b}{V} - \frac{Rc}{T^2} - \frac{RB_o c}{VT^2} + \frac{RB_o bc}{V^2 T^2} - A_o + \frac{A_o a}{V} + \frac{A_o d}{V^2}$$

Now our goal was to get the right side of the equation in terms of Z by using the equation $Z = \frac{PV}{RT}$. To do this we started by dividing both sides of the equation by RT and got $\frac{PV^2}{RT} = V + B_o - \frac{B_o b}{V} - \frac{c}{T^3} - \frac{B_o c}{VT^3} + \frac{B_o bc}{V^2 T^3} - \frac{A_o}{RT} + \frac{A_o a}{VRT} + \frac{A_o d}{V^2 RT}$

We can start simplifying the equation by factoring out common V factors from the terms which left us with this form

$$\frac{PV^2}{RT} = V + \left(B_o - \frac{A_o}{RT} - \frac{c}{T^3} \right) + \frac{1}{V} \left(-B_o b - \frac{B_o c}{T^3} + \frac{A_o a}{RT} \right) + \frac{1}{V^2} \left(\frac{B_o bc}{T^3} + \frac{A_o d}{RT} \right)$$

To get the form $\frac{PV}{RT}$ which is equal to Z we need to divide both sides of the equation by V which will allow us to express the equation as a Z function then we used the fact that Volume is the inverse of density to express the whole function in terms of ρ and Z, which was the intended purpose of all the manipulations of the original virial equation. Our final equation becomes

$$Z = 1 + \rho \left(B_o - \frac{A_o}{RT} - \frac{c}{T^3} \right) + \rho^2 \left(-B_o b - \frac{B_o c}{T^3} + \frac{A_o a}{RT} \right) + \rho^3 \left(\frac{B_o bc}{T^3} + \frac{A_o d}{RT} \right)$$

Where $\beta = B_o - \frac{A_o}{RT} - \frac{c}{T^3}$, $\gamma = -B_o b - \frac{B_o c}{T^3} + \frac{A_o a}{RT}$, and $\delta = \frac{B_o bc}{T^3} + \frac{A_o d}{RT}$

Now we have our virial equation in terms of $Z = Z(\rho, T)$. The next part of problem one requires us to come up with a methodology to find the constant d in our virial equation which is zero for pressures between 0 and 400 atm but we want to find the optimal d value at high pressures so we can calculate our Z value accurately for high pressures. To start off we were provided with the following experimental data and also, we have included a screenshot of our vba code for finding Z based on the constant d:

Experimental Data for Nitrogen										Deming and Shupe constants for the Beattie-Bridgeman Virial Equation of State				d optimal
Z-values	Temperature (deg C)					P-values, kmol/m ³	Temperature (deg C)					Ao	1528.600000	1528.600000
P (atm)	0	25	50	75	100	P (atm)	0	25	50	75	100	a	0.274800	0.274800
0	1.000	1.000	1.000	1.000	1.000	0	0	0	0	0	0	Bo	1.643000	1.643000
1	1.000	1.000	1.000	1.000	1.000	1	0.044615033	0.04087404	0.0377119	0.035	0.03266	b	-0.923500	-0.923500
3	0.999	0.999	1.000	1.000	1.001	3	0.133979079	0.12274488	0.1131357	0.10501	0.09788	c	2.200000E+06	#####
5	0.998	0.999	1.000	1.001	1.001	5	0.223522211	0.2045748	0.1885594	0.17484	0.16313	d	0.000000	1.413176409
10	0.996	0.998	1.000	1.001	1.002	10	0.447942102	0.40955958	0.3771189	0.34969	0.32594	R, cm3 atm / g k		
30	0.959	0.996	1.001	1.005	1.007	30	1.395673621	1.23114592	1.1302264	1.04489	0.97295	2.929247135		
50	0.958	0.996	1.004	1.009	1.014	50	2.328550804	2.05190987	1.878082	1.73458	1.61039			
100	0.994	1.004	1.018	1.027	1.033	100	4.53404811	4.07111997	3.7045076	3.40836	3.16154			
200	1.036	1.057	1.072	1.082	1.089	200	8.612940812	7.73397247	7.0357999	6.47022	5.99793			
300	1.134	1.146	1.154	1.160	1.163	300	11.80291889	10.7000117	9.8037835	9.05272	8.42443			
400	1.256	1.254	1.253	1.250	1.247	400	14.20860936	13.0379727	12.038911	11.2012	10.4759			
600	1.524	1.495	1.471	1.450	1.432	600	17.56497378	16.4042988	15.382143	14.4844	13.6838			
800	1.798	1.723	1.697	1.658	1.625	800	19.85096036	18.9780822	17.778144	16.8897	16.0781			
1000	2.067	1.988	1.923	1.867	1.819	1000	21.58443803	20.5603846	19.610966	18.7487	17.9542			

```

Function Z_BB2(rho As Double, T As Double, Param As Range, R As Double)

'Calculate the compressibility factor given density and Temp
'Rho [=] molar density, kmol/m3
'T [=] temperature deg C
'Param [=] excel range that contains the numerical constant of the BB eqn
'Z_BB [=] compressibility factor (calculated)

Dim Tk As Double
Tk = T + 273.15 'Temp in K

'Now retrieve constant of BB virial eqn

Dim a As Double, Ao As Double, b As Double, Bo As Double, c As Double, d As Double
Ao = Param.Cells(1, 1)
a = Param.Cells(2, 1)
Bo = Param.Cells(3, 1)
b = Param.Cells(4, 1)
c = Param.Cells(5, 1)
d = Param.Cells(6, 1)

'Now calculate beta, gamma, and delta based off derivations

Dim beta As Double, gamma As Double, delta As Double
beta = Bo - (Ao / (R * Tk)) - (c / Tk ^ 3)
gamma = -(Bo * b) - ((Bo * c) / Tk ^ 3) + ((Ao * a) / (R * Tk))
delta = ((Bo * b * c) / Tk ^ 3) + ((Ao * d) / (R * Tk))

'Now Calc Z
Z_BB2 = 1 + beta * rho + gamma * rho ^ 2 + delta * rho ^ 3
End Function

```

This function allows for the evaluation of the compressibility factor of nitrogen (or other gases described by the same parameter set) at a given temperature and density. By implementing this function in VBA, the user can easily compute Z values across a range of densities and temperatures directly within Excel, enabling automated data analysis and thermodynamic property estimation. The VBA function is designed to calculate the compressibility factor, Z, from specified values of density (ρ) and temperature (T) using the virial equation of state. The function takes as inputs the virial parameters (A_0 , a , B_0 , b , c , and d), the universal gas constant R , and the temperature and density of the system. Then we define another variable T_k , which helps us calculate the temperature in Kelvin because when we call our temperature in the excel sheet all the experimental data has temperature in Celsius, but our calculations require Kelvin, so our function converts it. Now to correctly call the constants for our function, we must dim them as variables and then we reference them using `param.cells(row,column)`. Now that we collect the constants in the excel function, the values will be assigned to each constant correctly. Finally, we defined beta, gama and delta as variables and inputted the equations needed to calculate those values based on the Z equation we derived above. Finally, the function computes Z by substituting these variables into the modified virial equation, in terms of beta, gamma, and delta we previously derived. By coding this calculation into VBA, the function allows for efficient evaluation of Z across multiple data points directly in Excel, facilitating the modeling of experimental pressure–density–temperature (P– ρ –T) data and comparison between theoretical and experimental results.

Now that we have the Z function coded into VBA we can use our excel file to calculate Z values for us. When called the Z function requires the user to input a rho in g/cm³, Temperature in Celsius, the given parameters, and our R value which requires units of cm³ atm/ g K. In the excel

template we were given a density table of values, but they are in units of kmol/m³ and our function requires g/cm³ for a correct output of Z. We created a new table using the function $\rho = \frac{P}{RTZ}$, using the Z values from the original table in the template, the T values also from the given table. For our R value we used the ideal gas constant $82.057 \frac{\text{cm}^3 \text{ atm}}{\text{mol K}}$ and to convert it we divided by the molecular weight of nitrogen gas and got $2.929 \frac{\text{cm}^3 \text{ atm}}{\text{g K}}$

ρ -values, g/cm ³		Temperature (deg C)				
P (atm)		0	25	50	75	100
0		0	0	0	0	0
1		0.00125	0.00115	0.00106	0.00098	0.00091
3		0.00375	0.00344	0.00317	0.00294	0.00274
5		0.00626	0.00573	0.00528	0.0049	0.00457
10		0.01255	0.01147	0.01056	0.0098	0.00913
30		0.0391	0.03449	0.03166	0.02927	0.02726
50		0.06523	0.05748	0.05261	0.04859	0.04511
100		0.12701	0.11404	0.10377	0.09548	0.08856
200		0.24128	0.21665	0.19709	0.18125	0.16802
300		0.33064	0.29974	0.27463	0.2536	0.23599
400		0.39803	0.36523	0.33725	0.31378	0.29346
600		0.49205	0.45954	0.4309	0.40575	0.38333
800		0.55609	0.53164	0.49802	0.47313	0.4504
1000		0.60465	0.57596	0.54936	0.52521	0.50295

Now that we have the correct rho values and R constant, we tested the Z function by creating a table of values when d = 0 and we got the following values of Z at various pressures and temperatures:

Z-values		Temperature (deg C)				
P (atm)		0	25	50	75	100
0		1	1	1	1	1
1		0.999533733	0.99978458	0.9999629	1.00009	1.00019
3		0.998617314	0.99936778	0.9999012	1.00029	1.00057
5		0.99772248	0.99897074	0.999856	1.00049	1.00096
10		0.995582688	0.99806193	0.9998154	1.00108	1.002
30		0.988163772	0.9956474	1.0006823	1.00427	1.00687
50		0.983402265	0.99519102	1.0031679	1.00881	1.01284
100		0.982069042	1.0023301	1.0160101	1.02552	1.03223
200		1.015688764	1.04489086	1.0639592	1.07676	1.08539
300		1.073833038	1.10685707	1.1276079	1.14059	1.1489
400		1.135704221	1.17273927	1.1945705	1.20793	1.21549
600		1.247294017	1.29352243	1.3203542	1.33554	1.34308
800		1.339775575	1.40615762	1.4291542	1.44782	1.45684
1000		1.418621703	1.48399292	1.5227676	1.54537	1.55695

The whole point of this problem is to find d which is done by minimizing the difference between the experimental pressures and model pressures. The model pressures are the given values we are given in the tables (0-1000 atm). To find P experimental we need to find all the Z values at d optimal and use the equation $Z = \frac{P}{RT\rho}$ which can be manipulated to $P = RT\rho Z$, but before find P experimental we need to find Z.

We found Z optimal by choosing a guess for D optimal which we started out by saying d optimal was one, and all the other constants in the equation remained the same. Then we created a new table for Z values at d optimal using our Z_BB function we coded in VBA. Below is the table once we found d optimal:

d optimal					
Z-values	Temperature (deg C)				
P (atm)	0	25	50	75	100
0	1	1	1	1	1
1	0.999533739	0.99978459	0.9999629	1.00009	1.00019
3	0.998617457	0.99936788	0.9999013	1.00029	1.00057
5	0.997723142	0.9989712	0.9998563	1.00049	1.00096
10	0.995588023	0.99806567	0.9998181	1.00108	1.002
30	0.98832512	0.99574886	1.0007547	1.00432	1.00691
50	0.984151588	0.99566076	1.0035002	1.00906	1.01302
100	0.987600891	1.00599886	1.0185605	1.02736	1.0336
200	1.053608584	1.07004365	1.0814314	1.08937	1.09476
300	1.171417382	1.17346578	1.1748784	1.17514	1.17488
400	1.305946178	1.29324522	1.2821033	1.27337	1.26543
600	1.568922422	1.53354456	1.5029368	1.47704	1.45439
800	1.804030421	1.77780856	1.7110368	1.67216	1.6374
1000	2.01542899	1.95656926	1.9011282	1.85225	1.80839

Now that Z optimal values were obtained, we used the equation $P = RT\rho Z$ to find all the experimental P values at the calculated Z values all based on d optimal. Which gave us this table:

P at optimal d					
	Temperature (deg C)				
	0	25	50	75	100
	0	0	0	0	0
	0.999533739	0.99978459	0.9999629	1.00009	1.00019
	2.998851221	3.00110473	2.9997038	3.00086	2.99871
	4.998612938	4.99985586	4.9992817	4.99748	4.99981
	9.995863682	10.000658	9.9981808	10.0008	10
	30.91736558	29.9924356	29.992648	29.9798	29.9973
	51.36490544	49.9829696	49.975111	50.0028	49.9519
	100.3659442	100.19909	100.05506	100.035	100.059
	203.3993406	202.468051	201.7596	201.363	201.058
	309.8987783	307.189996	305.42765	303.915	303.064
	415.906426	412.518412	409.29074	407.477	405.912
	617.6859928	615.46939	613.02657	611.188	609.383
	802.6831682	825.447968	806.61721	806.833	806.106
	975.0503096	984.189768	988.62619	992.099	994.164

Next, we used the QSUMMS function in the calculator to find the sum of the squared differences of model Pressures and experimental Pressures to be able to minimize the sums of those differences by manipulating d.

Cell Styles v

Styles

BASE

$\sum x^2$

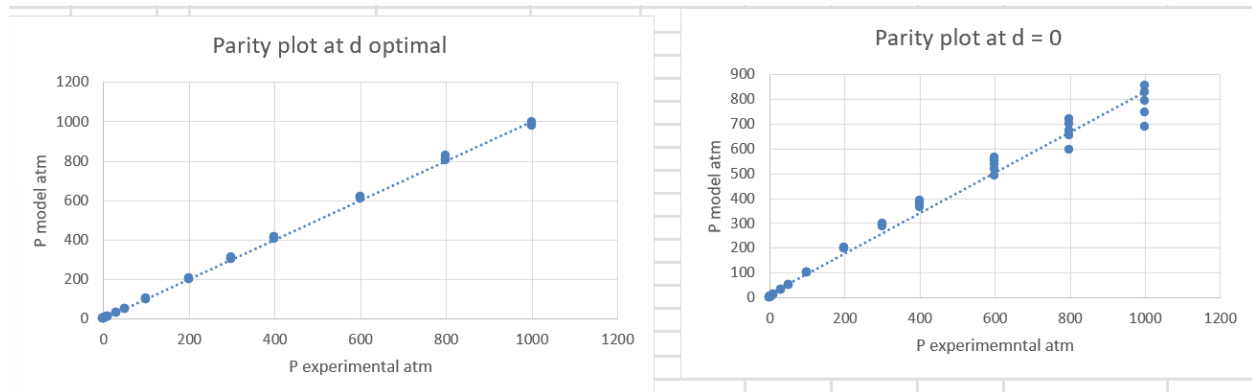
=SUMSQ(I42:I55-J60:N73)

H	I	J	K	L	M	N	O	P	Q
	1000	1.4186217103	1.49599292	1.5227676	1.54933	1.56699			
d optimal									
	Z-values	Temperature (deg C)							
	P (atm)	0	25	50	75	100			
	0	1	1	1	1	1			
	1	0.999533739	0.99978459	0.9999629	1.00009	1.00019			
	3	0.998617457	0.99936788	0.9999013	1.00029	1.00057			
	5	0.997723142	0.9989712	0.9998563	1.00049	1.00096			
	10	0.995588023	0.99806567	0.9998181	1.00108	1.002			
	30	0.98832512	0.99574886	1.0007547	1.00432	1.00691			
	50	0.984151588	0.99566076	1.0035002	1.00906	1.01302			
	100	0.987600891	1.00599886	1.0185605	1.02736	1.0336			
	200	1.053608584	1.07004365	1.0814314	1.08937	1.09476			
	300	1.171417382	1.17346578	1.1748784	1.17514	1.17488			
	400	1.305946178	1.29324522	1.2821033	1.27337	1.26543			
	600	1.568922422	1.53354456	1.5029368	1.47704	1.45439			
	800	1.804030421	1.77780856	1.7110368	1.67216	1.6374			
	1000	2.01542899	1.95656926	1.9011282	1.85229	1.80639			
P at optimal d									
		Temperature (deg C)							
		0	25	50	75	100	J60:N73		
		0	0	0	0	0			
		0.999533739	0.99978459	0.9999629	1.00009	1.00019			
		2.998851221	3.00110473	2.9997038	3.00086	2.99871			
		4.998612938	4.99858586	4.9982817	4.99748	4.99681			
		9.995863682	10.000658	9.9981808	10.0008	10			
		30.91736558	29.9924356	29.992648	29.99798	29.9973			
		36.913490544	49.9829686	48.975111	50.00028	49.9519			
		100.3658442	100.19909	100.05508	100.035	100.058			
		203.39934064	202.468051	201.7596	201.363	201.058			
		309.8987783	307.189996	305.42765	303.915	303.064			
		415.906426	412.518412	409.29074	407.477	405.912			
		617.6859928	615.46939	613.02657	611.188	609.383			
		802.6831682	825.447968	806.61721	806.833	806.106			
		975.0503096	984.189768	986.62619	992.099	994.164			

Now we were able to use solver in excel to minimize the value of the SUMSQ function by changing to d optimal which gave us a **d optimal value of 1.413**

Deming and Shupe constants for the Beattie-Bridgeman Virial Equation of State		d optimal
Ao	1528.600000	1528.600000
a	0.274800	0.274800
Bo	1.643000	1.643000
b	-0.923500	-0.923500
c	2.200000E+06	2.200000E+06
d	0.000000	1.413176409

Now that we were able to obtain d optimal, our next step was to create two parity plots of model Pressure versus experimental pressure at d equals zero and d equals its optimal value (1.413). For d = 0 we found the experimental pressures using the same equation as we previously did for d optimal. The two plots can be seen below:



What can be gathered from the graphs is that adding the d optimal value to the virial equation does help gather more accurate results because as can be seen at pressures above 400 atm when d equals zero the graph is less linear but the data is extremely linear when we included optimal for higher pressures. This tells us that calculating d and adding it to the virial equations gives us more accurate values for Z .

Problem 2

- Calculate the thermal properties of Nitrogen $\{\beta, C_p, C_v, \mu_{JT}, v_{\text{sound}}\}$ at temperature $T = -60^\circ\text{C}$ and pressure $P = 800\text{ atm}$ using the modified virial equation. Express your answers in **SI** units. Compare your results with the values obtained using the **(RKS)** Redlich-Kwong-Soave Equation of State and the values reported in the NIST Webbook.

In problem 2, we are asked to calculate the secondary thermal properties based on the virial equation we derived and then compare those values to RKS EOS and the experimental values from the NIST Website. To find the equations to derive these properties we went to the textbook and found the following equations:

$$\beta = \frac{1}{\rho} \cdot \frac{\left(\frac{\partial P}{\partial T}\right)_\rho}{\left(\frac{\partial P}{\partial \rho}\right)_T}$$

$$C_p - C_v = \frac{T}{\rho^2} \cdot \frac{\left(\frac{\partial P}{\partial T}\right)_\rho^2}{\left(\frac{\partial P}{\partial \rho}\right)_T} \quad \text{and therefore} \quad C_p = \frac{T}{\rho^2} \cdot \frac{\left(\frac{\partial P}{\partial T}\right)_\rho^2}{\left(\frac{\partial P}{\partial \rho}\right)_T} + C_v$$

$$C_v = C_v^{IG} + C_v^R \quad \text{and} \quad C_v^R = \int_{V=\infty}^V T \left(\frac{\partial^2 P}{\partial T^2}\right)_V dV$$

$$\mu_{JT} = \frac{\beta T - 1}{\beta c_p}$$

$$v^2 = \frac{1}{MW} \cdot \frac{C_p}{C_v} \cdot \left(\frac{\partial P}{\partial \rho} \right)_T$$

All of these equations are based on the derivative of Pressure with either changing temperature or density. But our virial equation we derived it in terms of Z, so first we found the derivatives of Z in terms of rho and temperature and then converted those derivatives into the form of Pressure. As a reminder our Z equation that we derived above is

$$Z = 1 + \rho \left(B_o - \frac{A_o}{RT} - \frac{c}{T^3} \right) + \rho^2 \left(-B_o b - \frac{B_o c}{T^3} + \frac{A_o a}{RT} \right) + \rho^3 \left(\frac{B_o b c}{T^3} + \frac{A_o d}{RT} \right)$$

The derivatives we need to find to use equations for the properties are $\left(\frac{\partial Z}{\partial T} \right)_\rho$, $\left(\frac{\partial Z}{\partial \rho} \right)_T$,

$\left(\frac{\partial^2 Z}{\partial T^2} \right)_\rho$ which can be derived from the equation above using the basic rules for finding

derivatives. Below, I have included the derivatives and grouped the terms for the derivatives in the form of beta, gamma, and delta.

$$\left(\frac{\partial Z}{\partial T} \right)_\rho = \rho \left(\frac{A_o}{RT^2} + \frac{3c}{T^4} \right) + \rho^2 \left(\frac{3B_o c}{T^4} - \frac{A_o a}{RT^2} \right) + \rho^3 \left(-\frac{3B_o b c}{T^4} - \frac{A_o d}{RT^2} \right)$$

The equations 3 term also can be grouped into $\rho\beta + \rho^2\gamma + \rho^3\delta$, using these groupings will later

come in handy when putting the derivatives into VBA functions.

$$\left(\frac{\partial Z}{\partial \rho} \right)_T = \left(B_o - \frac{A_o}{RT} - \frac{c}{T^3} \right) + 2\rho \left(-B_o b - \frac{B_o c}{T^3} + \frac{A_o a}{RT} \right) + 3\rho^2 \left(\frac{B_o b c}{T^3} + \frac{A_o d}{RT} \right)$$

In this equation in the vba function, the first set of parentheses will represent beta in the code, and the second set will represent gamma, and the third delta.

$$\left(\frac{\partial^2 Z}{\partial T^2} \right)_\rho = \rho \left(\frac{-2A_o}{RT^3} - \frac{12c}{T^5} \right) + \rho^2 \left(\frac{-12B_o c}{T^5} + \frac{2A_o a}{RT^3} \right) + \rho^3 \left(\frac{12B_o b c}{T^5} + \frac{2A_o d}{RT^3} \right)$$

This equation follows the same groupings as above with the 3 sets of parentheses in the code.

Now that all the derivatives of Z have been derived, the next step is to code them into vba so we can call and use the functions in our excel calculations. I have included screenshots of our vba code for the 3 derivatives below:

```
Function dZ_dT(rho As Double, T As Double, Param As Range, R As Double)
'Calculate the derivative of dZ/dT at constant rho based on equation of Z
'T [=] degrees Celsius
'Rho [=] molar density, kmol/m3
'Param [=] excel range that contains the numerical constant of the BB eqn

Dim Tk As Double
Tk = T + 273.15 'Temp in K

'Now retrieve the constants of BB virial eqn
Dim a As Double, Ao As Double, b As Double, Bo As Double, c As Double, d As Double
Ao = Param.Cells(1, 1)
a = Param.Cells(2, 1)
Bo = Param.Cells(3, 1)
b = Param.Cells(4, 1)
c = Param.Cells(5, 1)
d = Param.Cells(6, 1)

'Now calculate beta, gamma, and delta based off derivations

Dim beta As Double, gamma As Double, delta As Double
beta = (Ao / (R * Tk ^ 2)) + ((3 * c) / (Tk ^ 4))
gamma = ((3 * Bo * c) / (Tk ^ 4)) - ((Ao * a) / (R * Tk ^ 2))
delta = ((-3 * Bo * b * c) / (Tk ^ 4)) - ((Ao * d) / (R * Tk ^ 2))

'Now calculate dZ/dT
dZ_dT = (rho * beta) + (rho ^ 2 * gamma) + (rho ^ 3 * delta)

End Function
```

```
Function dZ_drho(rho As Double, T As Double, Param As Range, R As Double)
'Calculate the derivative of dZ/dT at constant rho based on equation of Z
'T [=] degrees Celsius
'Rho [=] molar density, kmol/m3
'Param [=] excel range that contains the numerical constant of the BB eqn

Dim Tk As Double
Tk = T + 273.15 'Temp in K

'Now retrieve the constants of BB virial eqn
Dim a As Double, Ao As Double, b As Double, Bo As Double, c As Double, d As Double
Ao = Param.Cells(1, 1)
a = Param.Cells(2, 1)
Bo = Param.Cells(3, 1)
b = Param.Cells(4, 1)
c = Param.Cells(5, 1)
d = Param.Cells(6, 1)

'Now calculate beta, gamma, and delta based off derivations

Dim beta As Double, gamma As Double, delta As Double
beta = (Bo - (Ao / (R * Tk))) - (c / Tk ^ 3)
gamma = ((-Bo * b) - ((Bo * c) / Tk ^ 3) + ((Ao * a) / (R * Tk)))
delta = (((Bo * b * c) / Tk ^ 3) + ((Ao * d) / (R * Tk)))

'Now calculate dZ/drho
dZ_drho = beta + (2 * rho * gamma) + (3 * rho ^ 2 * delta)

End Function
```

```

Function dZ2_dT2(rho As Double, T As Double, Param As Range, R As Double)
'Calculate the derivative of dZ/dT at constant rho based on equation of Z
'T [=] degrees Celsius
'Rho [=] molar density, kmol/m3
'Param [=] excel range that contains the numerical constant of the BB eqn

Dim Tk As Double
Tk = T + 273.15 'Temp in K

'Now retrieve the constants of BB virial eqn
Dim a As Double, Ao As Double, b As Double, Bo As Double, c As Double, d As Double
Ao = Param.Cells(1, 1)
a = Param.Cells(2, 1)
Bo = Param.Cells(3, 1)
b = Param.Cells(4, 1)
c = Param.Cells(5, 1)
d = Param.Cells(6, 1)

'Now calculate beta, gamma based off derivations
Dim beta As Double, gamma As Double, delta As Double
beta = ((-2 * Ao) / (R * Tk ^ 3)) - ((12 * c) / (Tk ^ 5))
gamma = ((-12 * Bo * c) / (Tk ^ 5)) + ((2 * Ao * a) / (R * Tk ^ 3))
delta = ((12 * Bo * b * c) / (Tk ^ 5)) + ((2 * Ao * d) / (R * Tk ^ 3))

'Now calculate dZ2/dT2
dZ2_dT2 = (rho * beta) + (rho ^ 2 * gamma) + (rho ^ 3 * delta)

End Function

```

All of the code for the three derivatives follows the format we used for the ZBB virial equation. We define the arguments of the function as rho for density, T for temperature, Param for the constants we are given, and R as the gas constant. All these values will be prompted by the user to enter when they call the function in excel. The next step in all the equations is to convert T to Tk because the user will input a temperature in Celsius and for our calculations to be accurate, we need the values in terms of Kelvin. The param.cells portion of the code will allow the user to select the constants from the table and input them into the function, referencing the correct constants in our equation. Then, in each individual function, we used my groupings of terms per the derivations above and we created the variables beta, gamma, and delta. From there we plugged the terms beta gamma and delta back into the overall derivatives with the rho variable and calculated the overall derivatives.

Our next step was to find the Z and density value of Nitrogen gas at the conditions of this problem which are Temperature of -60 degrees Celsius and pressure of 800 atm because Z and density are required to calculate the thermal properties. In this circumstance we are not given density or Z and they both depend on each other so to find both values we will set up a system of Equations for Z and then goal seek the difference between those two equations by changing density since Z depends on density. The two equations to calculate Z are the ZBB equation we derived and the general equation from the textbook $Z = \frac{P}{\rho RT}$.

I made a guess for density in g/cm³ and then in two separate cells I entered each equation for Z. For ZBB I called the ZBB function that we previously coded and then entered all the parameters with rho being our guess for density. Then for the second equation I inputted Z =

$$\frac{800 \text{ atm}}{\left(\text{guess} \frac{\text{g}}{\text{cm}^3}\right) \left(2.92 \frac{\text{atm cm}^3}{\text{g K}}\right) (213.15 \text{ K})}$$

In the next cell I subtracted the two Z equations from each other then used goal seek to change that cell to zero by changing density and I got the following values for nitrogen gas at -60 degrees C and 800 atm

$$\rho = 0.652 \frac{g}{cm^3} \text{ and } Z = 1.964$$

The next step to solving for the thermal properties was to find the derivatives above in terms of P instead of Z in the book and in class we were given the following relationships: $\left(\frac{\partial P}{\partial T}\right)_\rho =$

$$\rho R \left(Z + T \left(\frac{\partial Z}{\partial T} \right)_\rho \right)$$

$$\left(\frac{\partial P}{\partial \rho} \right)_T = RT \left(Z + \rho \left(\frac{\partial Z}{\partial \rho} \right)_T \right)$$

$$T \left(\frac{\partial^2 P}{\partial T^2} \right)_\rho = \rho R \left(2T \left(\frac{\partial Z}{\partial T} \right)_\rho + T^2 \left(\frac{\partial^2 Z}{\partial T^2} \right)_\rho \right)$$

Now we are ready to begin calculating our thermal properties. Below I have included a screenshot of our set up on excel:

QUESTION 2			
conditions at -60 C and 800 atm	-60 T in K	213.15	
Z using Z_BB	1.964246765	Z using	1.96424676
rho, g/cm3	0.652307829	goal seek for Z	-5.08721E-09
dZ/dT) rho	0.007311717		
dZ/drho) T	5.314311277		
dP/dT) rho	6.731147444	19.10482703	
dP/drho) T	3390.831812	9624.102843	
B, K^-1	0.003043198		
d2Z/dT2) rho	-0.000121786		
T*d2P/dT2) rho	-4.61665275		
CvR,	4.560707633		
CvIG, kJ/kmol K	21.04817817		
Cv, kJ/kmol K	25.60888581		
Cp - Cv	6.693482222	Cp - CV kJ/kmol K	18.99792289
Cp kJ/kmol K	44.60680869		
Joule thompson Coefficient K/bar	-0.012074708		
Speed of sound, m/s	598.4268954		

First, we started by calculating beta (thermal expansion coefficient) using the equation. $\beta = \frac{1}{\rho} \cdot$

$\left(\frac{\partial P}{\partial T}\right)_\rho$. To do this, we first used the Z derivative functions to call $\frac{\partial Z}{\partial T}$ and $\frac{\partial Z}{\partial \rho}$ into our excel file to

be used for our calculations. Once we calculated those values, we plugged them into the equations for the derivative of P (alongside the other necessary variables) we wrote above and

calculated $\frac{\partial P}{\partial T}$ and $\frac{\partial P}{\partial \rho}$. Then we plugged $\frac{\partial P}{\partial T}$ and $\frac{\partial P}{\partial \rho}$ and divided them by each other and multiplied by the inverse of our density value and got the following value:

Beta is 0.00304 K⁻¹

Next, we calculated C_V using the equation $C_V = C_V^{IG} + C_V^R$. The term C_V^{IG} since it is an ideal value, can be found using the CVIG function in the calculator. To find C_V^R we need to use the equation $C_V^R = \int_{V=\infty}^V T \left(\frac{\partial^2 P}{\partial T^2} \right)_V dV$ but for our problem we don't have volume instead we have density, but we can use their inverse relationship to get the integral in terms of density. Due to the inverse relationship, we can manipulate the equation into $C_V^R = - \int_{\rho=0}^{\rho} \frac{T}{\rho^2} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\rho} d\rho$. For this part, we kept messing up the derivation, so instead of P, we found the integral using our Z derivatives the equation became $C_V^R = - \int_{\rho=0}^{\rho} \frac{R}{\rho} \left(2T \left(\frac{\partial Z}{\partial T} \right)_{\rho} + T^2 \left(\frac{\partial^2 Z}{\partial T^2} \right)_{\rho} \right) d\rho$. Because the derivatives have rho in their functions, I simplified them back into terms of beta and gamma and delta to be able to undo the integral and then plug those values into a vba code to automate the process. Below the integral in terms of beta gamma and delta is written out β represents the beta terms for the first derivative β_2 represents the terms for the second derivatives:

$C_V^R = - \int_0^{\rho} \frac{R}{\rho} \left(2T(\rho\beta + \rho^2\gamma + \rho^3\delta) + T^2(\rho B_2 + \rho^2\gamma_2 + \rho^3\delta_2) \right) d\rho$, when undoing the integral, we ended up with the $C_V^R = -R \left(\left(2T \left(\beta\rho + \frac{\rho^2\gamma}{2} + \frac{\rho^3\delta}{3} \right) + T^2 \left(\rho B_2 + \frac{\rho^2\gamma_2}{2} + \frac{\rho^3\delta_2}{3} \right) \right) \right)$. Now this equation I put it into a vba code to be called and calculate the integral simply.

```

Function Integral(rho As Double, T As Double, Param As Range, R As Double)

Dim Tk As Double
Tk = T + 273.15 'Temp in K

Dim a As Double, Ao As Double, b As Double, Bo As Double, c As Double, d As Double
Ao = Param.Cells(1, 1)
a = Param.Cells(2, 1)
Bo = Param.Cells(3, 1)
b = Param.Cells(4, 1)
c = Param.Cells(5, 1)
d = Param.Cells(6, 1)

Dim beta As Double, gamma As Double, delta As Double
beta = (Ao / (R * Tk ^ 2)) + ((3 * c) / (Tk ^ 4))
gamma = ((3 * Bo * c) / (Tk ^ 4)) - ((Ao * a) / (R * Tk ^ 2))
delta = ((-3 * Bo * b * c) / (Tk ^ 4)) - ((Ao * d) / (R * Tk ^ 2))

Dim beta2 As Double, gamma2 As Double, delta2 As Double
beta2 = ((-2 * Ao) / (R * Tk ^ 3)) - ((12 * c) / (Tk ^ 5))
gamma2 = ((-12 * Bo * c) / (Tk ^ 5)) + ((2 * Ao * a) / (R * Tk ^ 3))
delta2 = ((12 * Bo * b * c) / (Tk ^ 5)) + ((2 * Ao * d) / (R * Tk ^ 3))

Dim term1 As Double, term2 As Double
term1 = (2 * Tk * ((beta * rho) + (gamma * rho ^ 2 * 0.5) + (rho ^ 3 * delta * (1 / 3))))
term2 = (Tk ^ 2 * ((rho * beta2) + (rho ^ 2 * gamma2 * 0.5) + (rho ^ 3 * delta2 * (1 / 3))))

Integral = -R * (term1 + term2)

End Function

```

I used the same format as my previous functions and copied and pasted my beta, gamma, and delta values from each of the previous derivatives and then inputted them into the general equation. This gave a CvR of 4.56 and then I used the CVIG function to calculate CVIG and added the two terms together and got:

Cv = 25.6 kJ/kmol K

The next step was to calculate Cp. To do this, first we calculated the equation $C_p - C_v = \frac{T}{\rho^2} \cdot$

$\frac{(\frac{\partial P}{\partial T})_\rho^2}{(\frac{\partial P}{\partial \rho})_T}$. We already previously calculated rho and the two derivatives, so we just plugged them and

got a Cp – Cv value of 18.99 kJ/kmol K and then to find Cp we added Cv to this term and got:

Cp = 44.6 kJ/kmol

The next thermal property we need to calculate is the Joule Thompson Coefficient using the equation $\mu_{JT} = \frac{\beta T - 1}{\beta C_p}$. We previously solved for beta and Cp so we simply plugged them into the equation and got

JT Coefficient = -0.012 K/bar

The final thermal property we need to calculate is the speed of sound using the equation $v^2 = \frac{1}{MW} \cdot \frac{C_p}{C_v} \cdot \left(\frac{\partial P}{\partial \rho} \right)_T$. And once again we have solved all these values, so we just plugged them into the equation and got

Speed of Sound is 598.42 m/s

Now that we have the values of the thermal properties from the virial equation, our next step is to compare these values to the values we obtain from the RKS virial equation, and the experimental values given to us by NIST website.

For the RKS virial equation we changed the data base to RKS and then used the Zcomp function to gather our Z values from the given conditions of temperature and pressure, we can use the Zcomp function because the calculator has the conditions for RKS already built in so it will accurately calculate Z. The book gives us the thermal properties in terms of the constants below:

$$\beta = \frac{1}{T} \frac{C_1}{C_2}$$

$$C_V R = R D_2 \frac{A}{B} \ln \left(1 + \frac{B}{Z} \right) \text{ and } C_V = C_V R + C_V I G$$

$$C_p - C_v = R \frac{C_1^2}{C_2}$$

$$\mu_{JT} = \frac{\beta T - 1}{\rho C_p}$$

$$v^2 = \frac{RT}{MW} \frac{C_p}{C_v} C_2$$

To find A, B, a, b, and lambda 1 and 2 we used the following derivations and equations from class



RKS EOS

$$P + \frac{a\sigma}{V(V+b)} = \frac{RT}{V-b}$$

$$\sigma = [1 + k(\omega)(1 - \sqrt{T_r})]^2$$

$$k(\omega) = 0.98509 + 1.5517\omega - 0.1013\omega^2$$

Peng-Robinson EOS

$$P + \frac{a\sigma}{V(V+b) + b(V-b)} = \frac{RT}{V-b}$$

$$\sigma = [1 + k(\omega)(1 - \sqrt{T_r})]^2$$

$k(\omega)$ = different #

Calculations of thermal properties w/ RKS

$$P + \frac{a\sigma}{V(V+b)} = \frac{RT}{V-b} \quad \left| \frac{V}{RT} \right. \quad a = \lambda_1 \left(\frac{RT_c}{P_c} \right)^2 \quad \lambda_1 \text{ to be determined}$$

$$b = \lambda_2 \frac{RT_c}{P_c} \quad \lambda_2 \text{ TBD}$$

$$\frac{PV}{RT} + \frac{a\sigma V}{V(V+b)RT} = \frac{V}{V-b}$$

$$Z = \frac{PV}{RT}$$

$$A = \frac{a\sigma P}{(RT)^2} \rightarrow A = \lambda_1 \frac{\sigma(T_r)P_r}{T_r^2}$$

$$B = \frac{bP}{RT} \rightarrow B = \lambda_2 \frac{P_r}{T_r}$$

we will put it in the format $Z = Z(A, B)$

we will put it in the format

$$Z + \frac{A\sigma}{VRT(1+\frac{B}{Z})} = \frac{1}{1-\frac{B}{Z}}$$

$$\frac{b}{V} = \frac{B}{Z}$$

$$\frac{a\sigma}{VRT} = \frac{A}{Z}$$

$$Z + \frac{A}{1+\frac{B}{Z}} = \frac{1}{1-\frac{B}{Z}}$$

$$\rightarrow Z + \frac{A}{B+Z} = \frac{Z}{Z-B} \rightarrow$$

$$Z^3 - Z^2 + Z(A-B-B^2) - AB = 0 \quad \text{RKS}$$

$$Z^3 - Z^2(1+B) + AZ - AB = 0 \quad \text{VOW}$$

$$A = \lambda_1 \frac{\sigma(T_r)P_r}{T_r^2} \quad B = \lambda_2 \frac{P_r}{T_r}$$

$$A_c = \lambda_1 \quad B_c = \lambda_2 \quad @ \text{ critical point}$$

$$Z^3 - Z^2 + Z(\lambda_1 - \lambda_2 - \lambda_2^2) - \lambda_1 \lambda_2 = 0$$

Because all the roots @ critical point are equal to Z_c

$$Z^3 - Z^2 + Z(\lambda_1 - \lambda_2 - \lambda_2^2) - \lambda_1 \lambda_2 = (Z - Z_c)^3 = Z^3 - 3Z^2Z_c + 3ZZ_c^2 - Z_c^3$$

$$z^3 - 3z^2z_c + 3zz_c^2 - z_c^3$$

$$-1 = -3z_c \rightarrow z_c = \frac{1}{3}$$

$$\lambda_1 - \lambda_2 - \lambda_2^2 = 3z_c^2 = 3 \cdot \frac{1}{9} = \frac{1}{3} \rightarrow$$

$$\lambda_1 \lambda_2 = z_c^3 = -\frac{1}{27}$$

$$\lambda_1 - \lambda_2 - \lambda_2^2 = \frac{1}{3}$$

Two Eqs
Two unknowns
use Newton method

$$\lambda_1 \lambda_2 = \frac{1}{27}$$

$$\lambda_1 = 0.42748$$

$$\lambda_2 = 0.08664$$

$$z_c = \frac{1}{3}$$

These derivations gave us the values for our constants for the RKS equation and then to calculate the C and D values we used the equations from the book:

$$D_1 = -K \sqrt{\frac{T_r}{\sigma}}$$

$$C_1 = Z + (1 - D_1) \frac{A}{B + Z}$$

$$C_2 = Z + \frac{BZ}{(Z - B)^2} - \frac{AZ}{(B + Z)^2}$$

$$D_2 = \frac{K(K+1)T_r}{2\sigma\sqrt{T_r}}$$

Then we plugged all of the constants into the equations above for our thermal properties and all of the values can be seen below:

RKS				
Z	1.934568583			
T, C	-60			
T, K	213.15			
Tc, K	126.2			
P atm	800			
Pc bar	33.943875			
R, atm cm ³ / g k	2.929247135			
omega	0.04			
k(omega)	0.54689824			
Tr	1.688985737			
Pr	23.88059701			
sigma	0.699136631			
lambda 1	0.42748			
lambda 2	0.08664			
a	13865.6892			
b	2.678246612			
A	2.501906491			
B	1.22500438			
C1	3.399520606			
C2	6.156665504			
D1	-0.850038288			
D2	0.78630169			
Beta K ⁻¹	0.002590519			
Cp kJ/kmol K	43.20535758			
Cv kJ/kmol K	27.598194	CvR	6.550015822	
JT K/bar	-0.04442118	rho	0.662314869	
Speed of sound m/S	609.7648702			

The results obtained from the virial equation and the Redlich–Kwong–Soave (RKS) equation show strong agreement, with both models yielding closely matching thermal property values.

Now we must compare our virial equation to the NIST data for the values. From NIST the thermal properties are:

Peng		Nist
Cp	43.1053	41.633
Cv	25.7043	25.082
JT	-0.02527	-0.08
Speed of sound	631.321	842.63
Beta	0.0035	
Z	1.76717	
Cv IG	21.0482	
Cv R	4.65614	

The table above compares the thermal properties obtained from the Peng–Robinson equation with the corresponding NIST reference values. The calculated heat capacities (C_p and C_v) and thermal expansion coefficient (β) show strong agreement with the NIST data. The Joule–Thomson (JT) coefficient exhibits a small deviation, with the NIST value at -0.08 k/bar and our calculated value at -0.01 k/bar. Similarly, the speed of sound differs slightly, with NIST reporting 842 m/s compared to our 598 m/s. However, our predicted value aligns closely with those obtained using both the RKS and Peng–Robinson equations, indicating good overall consistency among the models.

Problem 3

To calculate the power, we can use the setup for a compressor. We are given the inlet temperature and pressure $T_1 = -60^\circ\text{C}$, $P_1 = 810.6 \text{ bar}$ and the outlet pressure of $P_2 = 1013.25 \text{ bar}$. We will need to calculate the H^R and S^R values using the equations from the book.

$\ln \phi$	$(Z-1) - \ln Z + \int_0^P (Z-1) \frac{d\rho}{\rho}$
$H = H^{IG} + H^R$	$\frac{H^R}{RT} = (Z-1) - \int_0^P T \left(\frac{\partial Z}{\partial T} \right)_P \frac{d\rho}{\rho}$
$S = S^{IG} + S^R$	$\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi$

From the previous question we know what Z and $\frac{\partial Z}{\partial T}$ so we can use these functions to find H^R and S^R .

```

Function Enthalpy_R(rho As Double, Z As Double, T As Double, Param As Range, R As Double) As Double
    Dim Tk As Double
    Tk = T + 273.15 ' °C to K

    Dim Ao As Double, a As Double, Bo As Double, b As Double, c As Double, d As Double
    Ao = Param.Cells(1, 1).Value
    a = Param.Cells(2, 1).Value
    Bo = Param.Cells(3, 1).Value
    b = Param.Cells(4, 1).Value
    c = Param.Cells(5, 1).Value
    d = Param.Cells(6, 1).Value

    Dim beta_1 As Double, gamma_1 As Double, delta_1 As Double
    beta_1 = (Ao / (R * Tk ^ 2)) + ((3 * c) / (Tk ^ 4))
    gamma_1 = ((3 * Bo * c) / (Tk ^ 4)) - ((Ao * a) / (R * Tk ^ 2))
    delta_1 = ((-3 * Bo * b * c) / (Tk ^ 4)) - ((Ao * d) / (R * Tk ^ 2))

    ' Residual enthalpy (J/mol)
    Enthalpy_R = ((Z - 1) - Tk * (beta_1 * rho + (gamma_1 * rho ^ 2) / 2 + (delta_1 * rho ^ 3) / 3))
End Function

```

```

Function Entropy_R(rho As Double, Z As Double, T As Double, Param As Range, R As Double) As Double
    Dim Tk As Double
    Tk = T + 273.15

    Dim Ao As Double, a As Double, Bo As Double, b As Double, c As Double, d As Double
    Ao = Param.Cells(1, 1).Value
    a = Param.Cells(2, 1).Value
    Bo = Param.Cells(3, 1).Value
    b = Param.Cells(4, 1).Value
    c = Param.Cells(5, 1).Value
    d = Param.Cells(6, 1).Value

    Dim beta_1 As Double, gamma_1 As Double, delta_1 As Double
    beta_1 = (Ao / (R * Tk ^ 2)) + ((3 * c) / (Tk ^ 4))
    gamma_1 = ((3 * Bo * c) / (Tk ^ 4)) - ((Ao * a) / (R * Tk ^ 2))
    delta_1 = ((-3 * Bo * b * c) / (Tk ^ 4)) - ((Ao * d) / (R * Tk ^ 2))

    Dim HR As Double, lnPhi As Double

    lnPhi = (Z - 1) - Log(Z) + (beta_1 * rho + (gamma_1 * rho ^ 2) / 2 + (delta_1 * rho ^ 3) / 3 - Log(rho))
    ' simplified integral

    ' Residual entropy (J/mol·K)
    Entropy_R = Enthalpy_R(rho, Z, T, Param, R) - lnPhi
End Function

```

To find H^R as a dimensionless unit, we need to make a VBA code based on the equation given to us. The terms outside of the integral are constants while the function $\frac{\partial Z}{\partial T}$ is dependent on rho. The integral of this function comes out to be $\frac{\partial Z}{\partial T} = (\beta \cdot \rho + \rho^2 \cdot \gamma + \rho^3 \cdot \sigma)$ where beta, gamma, and delta are calculated the same as before. The other terms are included as constants. Our final equation is then $H^R = (Z - 1) - Tk \cdot (\beta \cdot \rho + \rho^2 \cdot \gamma + \rho^3 \cdot \sigma)$. For the residual entropy, we first have to calculate the fugacity using the equations given in the book. The Z values that aren't in the integral are constants while the Z in the integral is a function which simplifies to $Z - 1 = (\beta + \rho \cdot \gamma + \rho^2 \cdot \sigma) - 1$. After integrating, we get $Z - 1 = \left(\beta \cdot \rho + \frac{\rho^2}{2} \cdot \gamma + \frac{\rho^3}{3} \cdot \sigma \right) - \ln(\rho)$. So our final Fugacity equation is as written in the VBA function. We can then calculate S^R by calling a function that we previously done in our new function which is H^R to make things easier. In Excel we solve for $H = H^{IG} + H^R$ using the code we just found the same goes for the Entropy equation. Since we aren't given an outlet

temperature, we can initially make a guess for it then goal seek $S_1 = S_2$ by changing temperature. Then we can go ahead to solve the power by $H_2 - H_1$ then divide by molecular weight of nitrogen to get the power in mW.

Part 3					
	S1	S2rev			
Temp, C	-60	-54.47598465		Z for S2 Rev	
Press, bar	810.6	1013.25			
				Density	0.693517275
Z value	1.964238211	2.251072711		Z= P/rhoRT	2.251072711
molar enthalpy, kJ/kmol	-148.7221123	-126.1118316		Z using ZBB	2.235839098
molar entropy, kJ/molK	-2.831146116	-2.81296611		Z balance	0.015233613
			Entropy balance	0.018180005	
power kW	22.61028069				
Hr 1	-0.94802063				
Hr 2	-0.664873411				
Sr 1	-1.673360095				
Sr 2	-1.479280043				

To compare to RKS we have the same set up as part number 2

RKS						
	S1	S2				
temp C	-60	-25.7927	S1 Z	1.934568583	S2 Z	2.099242
Pres bar	810.6	1013.25	S1 rho	23.64301984	S2 rho	23.46904
Hr/RT	0.275785	-12.4981	Constants for S2			
Enthalpy kJ/kg	-71.302	-970.545	Z		2.099242091	
Sr/R	-1.19475	-1.49624	T, C		-25.79275	
Entropy kJ/kg K	-2.68912	-2.68912	T, K		247.35725	
			Te, K		126.2	
work kJ/kg	899.2429		P bar		1013.25	
Entropy balance	1.44E-07		Pc bar		33.943875	
			R, atm cm3 / g k		2.929247135	
			omega		0.04	
			k(omega)		0.54689824	
			Tr		0.175520585	
			Pr		29.85074627	
			sigma		1.736528986	
			lambda 1		0.42748	
			lambda 2		0.08664	
			a		0.649568806	
			b		2.67825E-05	
			A		27.0212946	
			B		0.004554888	
			C1		17.17652389	
			C2		-10.7148303	
			D1		-0.17387195	
			D2		0.102051706	

Problem 4

To verify whether the Beattie-Bridgeman Equation of State is consistent at the critical point of nitrogen we must use this equation from the book

$$\left(\frac{\partial P}{\partial \rho}\right)_T = RT \left(Z + \rho \left(\frac{\partial Z}{\partial \rho}\right)_T \right)$$

and write a VBA function for this and its second derivative.

We already have a function for dZ/drho so we can just call it in our VBA which then looks like this

```
Function dP_drho(rho As Double, T As Double, Param As Range, R As Double, Z As Double) As Double
    ' Calculate dP/drho for the Beattie-Bridgeman EOS
    ' T [°C], rho [mol/cm³], Param = constants range

    Dim Tk As Double
    Tk = T + 273.15    ' Temperature in K

    ' Retrieve Beattie-Bridgeman parameters
    Dim Ao As Double, a As Double, Bo As Double, b As Double, c As Double, d As Double
    Ao = Param.Cells(1, 1)
    a = Param.Cells(2, 1)
    Bo = Param.Cells(3, 1)
    b = Param.Cells(4, 1)
    c = Param.Cells(5, 1)
    d = Param.Cells(6, 1)

    ' Compute beta, gamma, delta
    Dim beta As Double, gamma As Double, delta As Double
    beta = Bo - (Ao / (R * Tk)) - (c / Tk ^ 3)
    gamma = -Bo * b - (Bo * c / Tk ^ 3) + (Ao * a / (R * Tk))
    delta = (Bo * b * c / Tk ^ 3) + (Ao * d / (R * Tk))

    ' Compute dZ/drho
    Dim dZdrho As Double
    dZdrho = beta + 2 * gamma * rho + 3 * delta * rho ^ 2

    ' Compute dP/drho = R*T*(Z + rho*dZ/drho)
    dP_drho = R * Tk * (Z + rho * dZdrho)
End Function
```

```

Function d2P_drho2(rho As Double, T As Double, Param As Range, R As Double, Z As Double)

'Calculate the derivative of dZ/dT at constant rho based on equation of Z
'T [=] degrees Celsius
'Rho [=] molar density,g/cm3
'Param [=] excel range that contains the numerical constant of the BB eqn

Dim Tk As Double
Tk = T + 273.15 'Temp in K

'Now retrieve the constants of BB virial eqn
Dim a As Double, Ao As Double, b As Double, Bo As Double, c As Double, d As Double
Ao = Param.Cells(1, 1)
a = Param.Cells(2, 1)
Bo = Param.Cells(3, 1)
b = Param.Cells(4, 1)
c = Param.Cells(5, 1)
d = Param.Cells(6, 1)

'Now calculate beta, gamma, and delta based off derivations

Dim beta As Double, gamma As Double, delta As Double
beta = Bo - (Ao / (R * Tk)) - (c / Tk ^ 3)
gamma = -Bo * b - (Bo * c / Tk ^ 3) + (Ao * a / (R * Tk))
delta = (Bo * b * c / Tk ^ 3) + (Ao * d / (R * Tk))
' Compute d2Z/drho2
Dim d2Zdrho2 As Double
d2Zdrho2 = 2 * gamma + 6 * delta * rho
' Compute dZ/drho
Dim dZdrho As Double
dZdrho = beta + 2 * gamma * rho + 3 * delta * rho ^ 2

'Now calculate dZ/drho
d2P_drho2 = R * Tk * (2 * (beta + 2 * dZdrho) + rho * d2Zdrho2)
End Function

```

When we call our functions in excel, we should use the critical temperature for nitrogen and if the values are equal or close to zero then the Beattie-Bridgeman Equation of State is consistent at the critical point of nitrogen. Our results were

1417.790827 dP/drho
5949.557602 d2P/drho2

When we tested the Beattie-Bridgeman (BB) equation of state at the critical point of nitrogen, it did not satisfy the critical conditions, meaning the predicted values for pressure, density, or compressibility did not match the expected critical behavior.

To correct this issue, an additional adjustable parameter can be introduced into the equation, such as a higher-order density term. Adding this parameter increases the flexibility of the model, allowing it to be recalibrated so that the predicted critical pressure, temperature, and density more closely match experimental values. In doing so, the modified equation can better capture the correct curvature and slope at the critical point, improving its overall consistency and accuracy.